

AI-Powered Customer Recommendation & Segmentation Engine

End-to-End Retail Analytics on the Online Retail II Dataset

Customer Segmentation • KNN Recommendation Engine • Market Basket Analysis • Customer Lifetime Value • Explainable AI

Metric	Value
Clean transactions	776,579
Customers	5,852
Products	4,620
Total revenue	GBP 17,068,583
Date range	2009-12-01 to 2011-12-09

A business-framed machine learning solution — every phase ends in a concrete commercial action.

1. The Dataset

Online Retail II is a real transactional dataset from a UK-based online gift retailer, covering roughly two years of sales (2009-12-01 to 2011-12-09). Unlike many tidy Kaggle datasets, it is genuine commercial data — complete with returns, cancellations, administrative charges and missing identifiers — which makes it ideal for demonstrating an end-to-end, production-minded workflow.

Column	Meaning
Invoice	Order number ('C' prefix = cancellation)
StockCode	Product ID
Description	Product name
Quantity	Units purchased (negative = return)
InvoiceDate	Purchase timestamp
Price	Unit price (GBP)
Customer ID	Customer identifier
Country	Customer country

The raw file holds **1,067,371** line items. The business questions we set out to answer: *Who are our most valuable customers? Who is about to churn? What should we recommend to each customer? Which products sell together? Where should marketing spend go?*

2. What Was Done — Methodology

The solution is a ten-phase Python pipeline. Each phase is a standalone, reproducible module orchestrated by `run_pipeline.py`:

Phase	Technique	Business output
2 Cleaning	Rule-based quality funnel	Trustworthy data
3 EDA	Visual + statistical analysis	Where revenue comes from
4 Feature Eng.	RFM + behavioural features	Customer intelligence
5 Segmentation	KMeans + DBSCAN	Targetable segments
6 Recommender	User-based KNN collab. filtering	Per-customer product recs
7 Market Basket	Apriori association rules	Bundle opportunities
8 CLV	RF / XGBoost / LightGBM regression	Future revenue per customer
9 Explainability	SHAP values	Why behind each score
10 BI + Report	Power BI star schema	Executive dashboard

3. Data Cleaning

Recommendations are only as good as the data behind them. We removed every record that would corrupt customer intelligence and report each dropped row in a transparent quality funnel:

Step	Rows
Raw rows	1,067,371
Duplicates removed	-34,335
Missing Customer ID	-235,151
Non-product codes	-3,660
Returns / cancellations	-17,586
Non-positive price	-60
Clean sale rows	776,579

Result: **776,579** clean sale rows across **5,852** customers and **4,620** products, totalling **GBP 17,068,583**. **Business value:** poor-quality data produces wrong recommendations and mis-targeted campaigns — this step protects every downstream decision.

4. Exploratory Data Analysis

Three lenses — customers, products and time — reveal where the business actually makes its money.

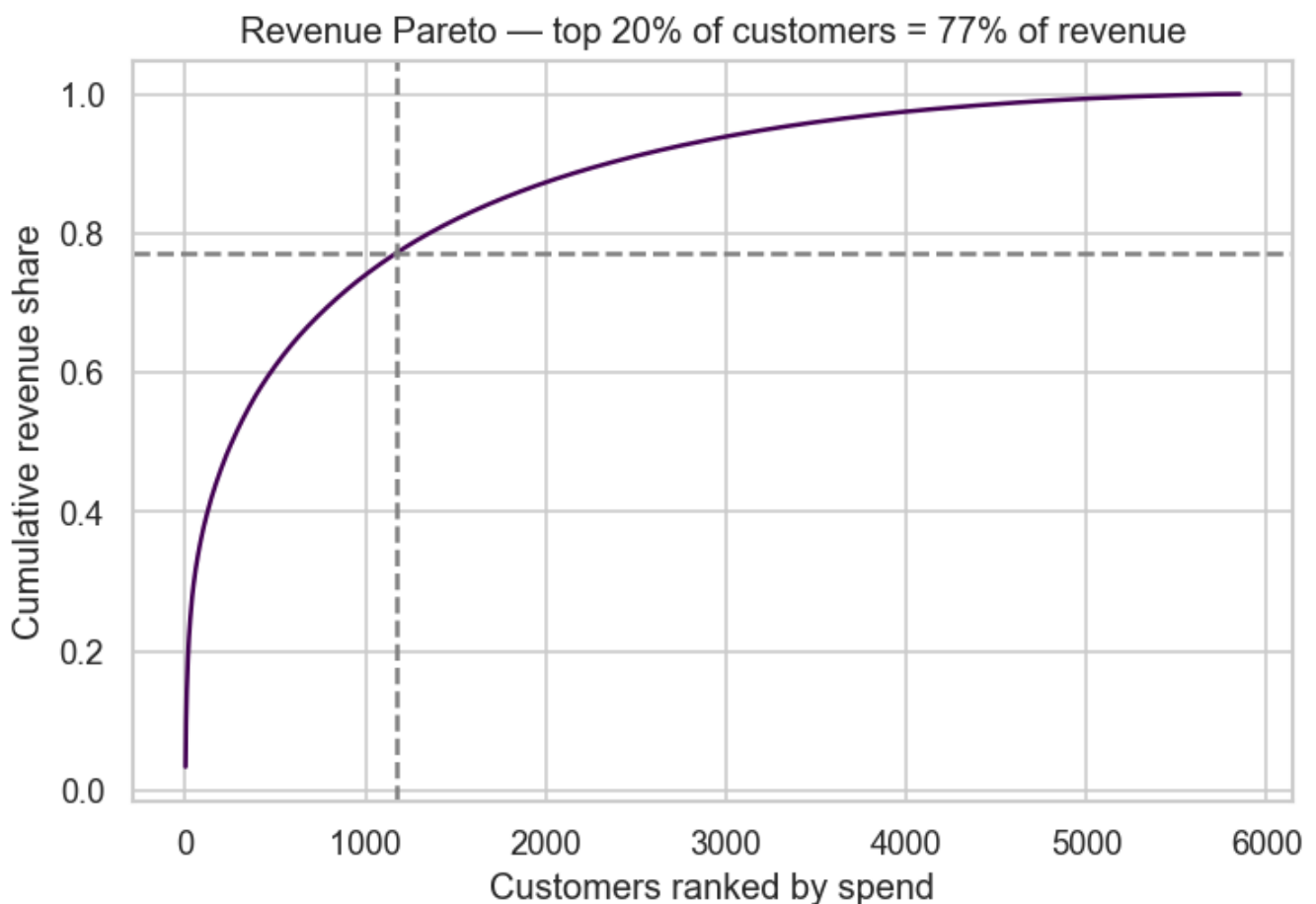


Figure 1. Revenue is highly concentrated: the top 20% of customers drive ~77% of revenue (Pareto principle in action).

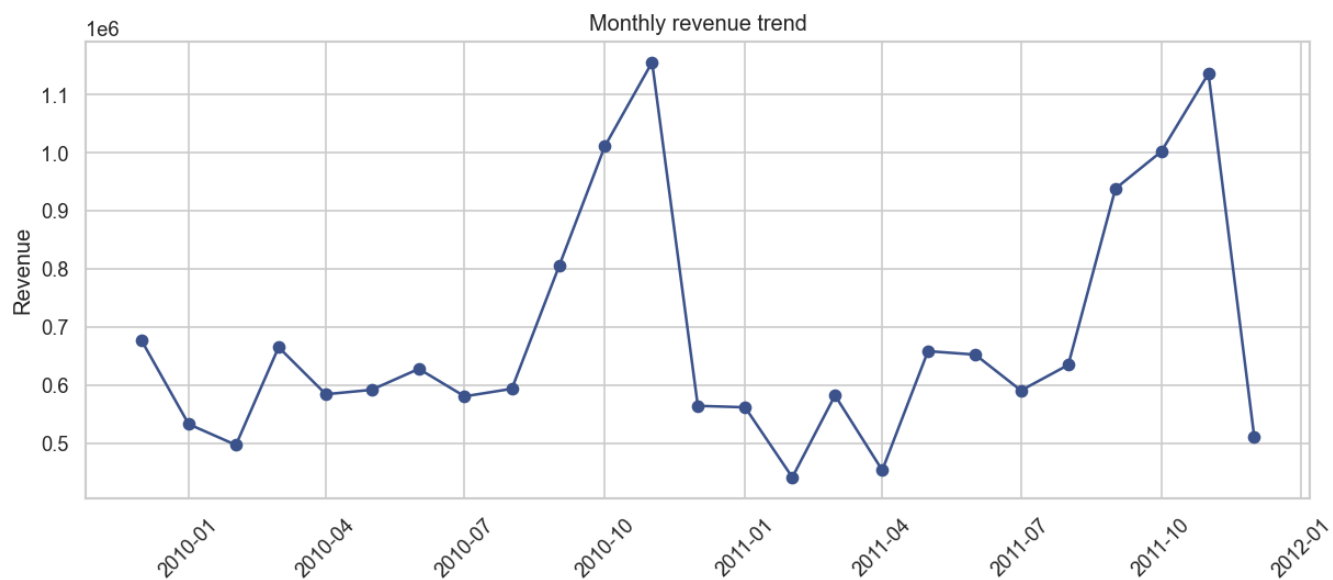


Figure 2. Monthly revenue trend — clear seasonal build-up toward the Q4 gifting peak, informing inventory and campaigns.

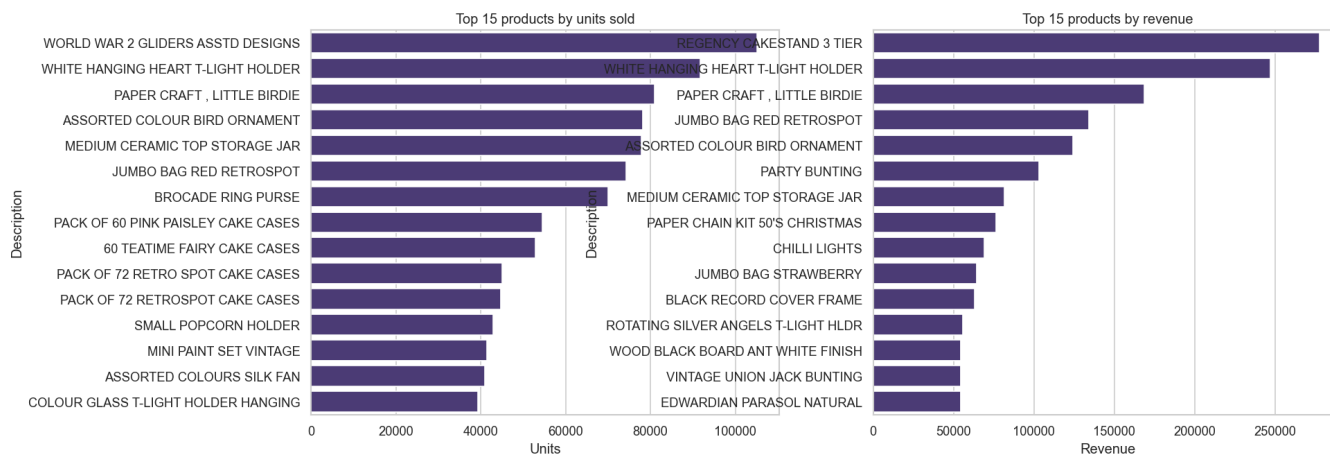


Figure 3. Top products by units and by revenue — a small set of SKUs (cakestands, t-light holders) dominates sales.

Business value: revenue concentration justifies a VIP-retention strategy; seasonality drives inventory and campaign timing; the long tail of rarely-sold products is a candidate for rationalisation.

5. Customer Segmentation

We engineered an **RFM** feature set — Recency (days since last purchase), Frequency (orders) and Monetary (total spend) — enriched with average basket value, product diversity, tenure and purchase cadence. RFM is heavily right-skewed, so features were log-transformed and standardised before clustering.

KMeans & DBSCAN

KMeans partitions every customer into business segments; K was chosen with the elbow and silhouette methods. **DBSCAN** adds a density view that isolates unusual (wholesale-scale) customers as noise.

Segment	Customers	Avg Recency	Avg Freq	Avg Monetary	Rev share
VIP	1,197	44d	18.9	GBP 10,831	76.0%
Loyal	1,606	58d	5.0	GBP 1,280	12.0%
Occasional	1,654	334d	2.3	GBP 1,061	10.3%
At-Risk	1,395	339d	1.6	GBP 210	1.7%

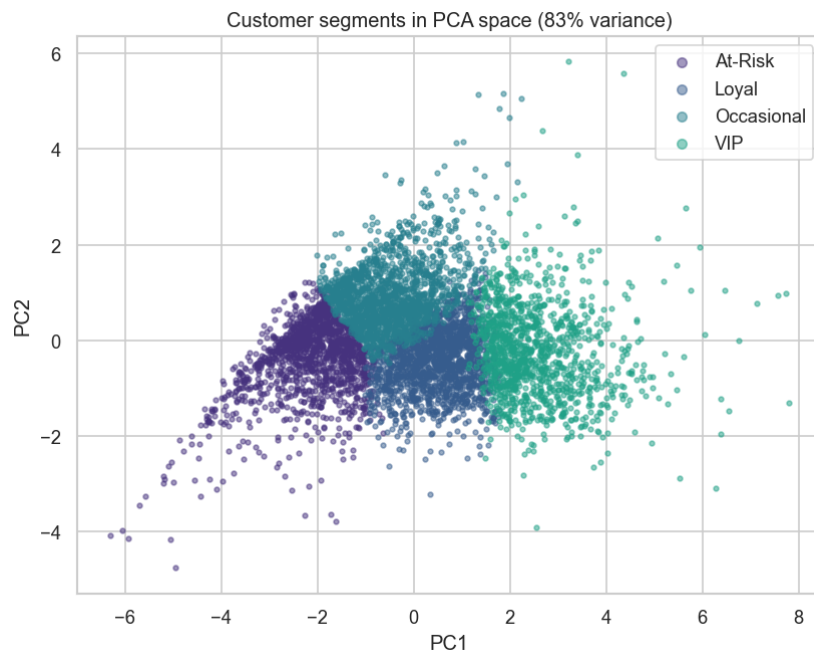


Figure 4. Customer segments projected into 2-D PCA space — well-separated, interpretable groups.

Segment playbook. VIP — protect and reward (loyalty perks, early access). Loyal — grow share of wallet via personalised cross-sell. Occasional — reactivate with cadence-timed win-back. At-Risk — low-cost automated reminders before churn.

6. KNN Recommendation Engine (Core)

The core deliverable. A sparse customer x product purchase matrix feeds a **user-based K-Nearest-Neighbours collaborative filter**: for a target customer we find the K most similar customers, pool the products they bought, score by similarity-weighted votes, drop items already owned, and return the top-10 — *'customers like you also bought...'*

Parameter Tuning

We tuned the **distance metric** (cosine, Euclidean, Manhattan) and **K** (3–15) against a leakage-free held-out evaluation: for each test customer 20% of their basket is hidden, recommendations are generated from the rest, and we measure how many hidden items are recovered. **Crucially, the customer's own row is excluded from the neighbour search** so metrics are not inflated by leakage.

Metric	K	Precision@10	Recall@10	Hit Rate@10	MAP@10
cosine	3	0.073	0.066	0.422	0.049
cosine	5	0.088	0.092	0.512	0.073
cosine	7	0.107	0.104	0.547	0.087
cosine	9	0.115	0.105	0.577	0.088
cosine	11	0.128	0.128	0.628	0.106
cosine	15	0.123	0.128	0.607	0.105

Best configuration: cosine distance, K = 11 — Hit Rate@10 = 62.8%, Precision@10 = 12.8%, MAP@10 = 0.106. Cosine clearly beats Euclidean/Manhattan on sparse binary purchase data.

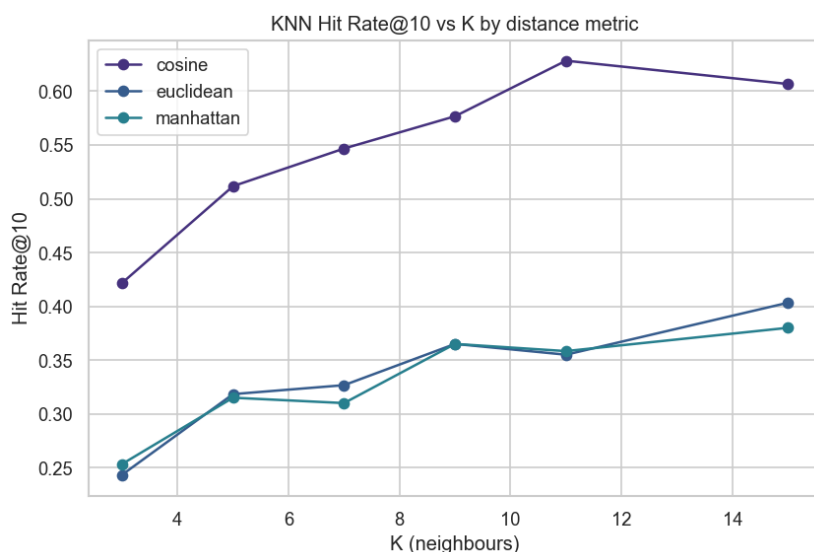


Figure 5. Hit Rate@10 vs K across the three distance metrics. Cosine dominates throughout.

An 'also-bought' item-similarity view (item-based KNN) complements the engine for product pages. Example associations:

Product	Also bought	Similarity
WHITE HANGING HEART T-LIGHT HOLDER	WHITE HANGING HEART T-LIGHT HOLDER	0.60
WHITE HANGING HEART T-LIGHT HOLDER	HEART OF WICKER LARGE	0.46
WHITE HANGING HEART T-LIGHT HOLDER	CANDELLHOLDER PINK HANGING HEART	0.45
WHITE HANGING HEART T-LIGHT HOLDER	HEART OF WICKER SMALL	0.44
WHITE HANGING HEART T-LIGHT HOLDER	CREAM HEART CARD HOLDER	0.41

Business value: a top-10 recommendation list is generated for **every** customer and exported for CRM/email activation — directly driving cross-sell and upsell (typically +5–15% basket revenue).

7. Market Basket Analysis (Apriori)

Where the recommender answers 'what should this customer buy?', market basket analysis answers 'which products sell together?' We built one basket per invoice (UK market, top 300 products) and mined **Apriori** association rules, ranked by **lift**:

- **Support** — how often the itemset appears.
- **Confidence** — $P(\text{consequent} | \text{antecedent})$.
- **Lift** — how much more likely the pair is than chance ($\text{lift} > 1$ = positive association).

If basket has	Also recommend	Conf.	Lift
PINK REGENCY TEACUP AND SAUCER	GREEN REGENCY TEACUP AND SAUCE	84%	26.9
GREEN REGENCY TEACUP AND SAUCER	PINK REGENCY TEACUP AND SAUCE	67%	26.9
ROSES REGENCY TEACUP AND SAUCER	GREEN REGENCY TEACUP AND SAUCE	72%	23.2
GREEN REGENCY TEACUP AND SAUCE	ROSES REGENCY TEACUP AND SAUCE	80%	23.2
TOILET METAL SIGN	BATHROOM METAL SIGN	78%	20.1
BATHROOM METAL SIGN	TOILET METAL SIGN	54%	20.1
ALARM CLOCK BAKELIKE RED	ALARM CLOCK BAKELIKE GREEN	61%	17.8
ALARM CLOCK BAKELIKE GREEN	ALARM CLOCK BAKELIKE RED	67%	17.8

Matching Regency teacup colours co-occur at a lift near **27x** — an extremely strong, actionable signal.

Business value: product bundling, store/site layout, and high-confidence basket upsell.

8. Customer Lifetime Value Prediction

We predict each customer's spend over the **next 6 months**. To avoid look-ahead leakage, features are built from a 12-month calibration window and the target is revenue in the following 6-month holdout window. The target is right-skewed with a spike at zero (churners), so models are trained on $\log_{1p}(\text{revenue})$ and evaluated on the original GBP scale.

Model	RMSE (GBP)	MAE (GBP)	R2
XGBoost	1,511	471	0.60
RandomForest	1,623	482	0.54
LightGBM	1,626	494	0.54

XGBoost wins with $R^2 = 0.60$ and RMSE of GBP 1,511 on out-of-sample customers — a strong result for noisy six-month-ahead revenue.

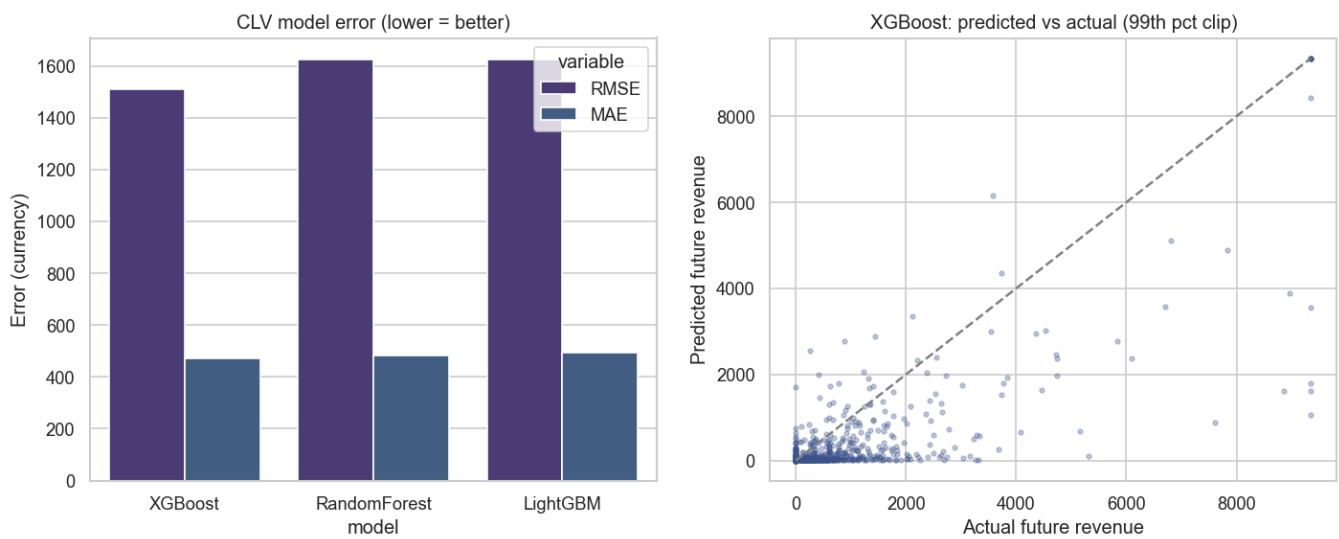


Figure 6. Model error comparison (left) and predicted vs actual future revenue for the best model (right).

9. Explainable AI (SHAP)

A CLV score the business cannot interpret is hard to act on. **SHAP** decomposes every prediction into per-feature contributions, answering both 'what drives future spend overall?' and 'why is this particular customer high-value?'

Rank	Feature	Mean SHAP
1	Monetary	0.699
2	Recency	0.464
3	Frequency	0.424
4	Tenure	0.363
5	AvgInterpurchase	0.257
6	ProductDiversity	0.244
7	AvgBasketValue	0.229

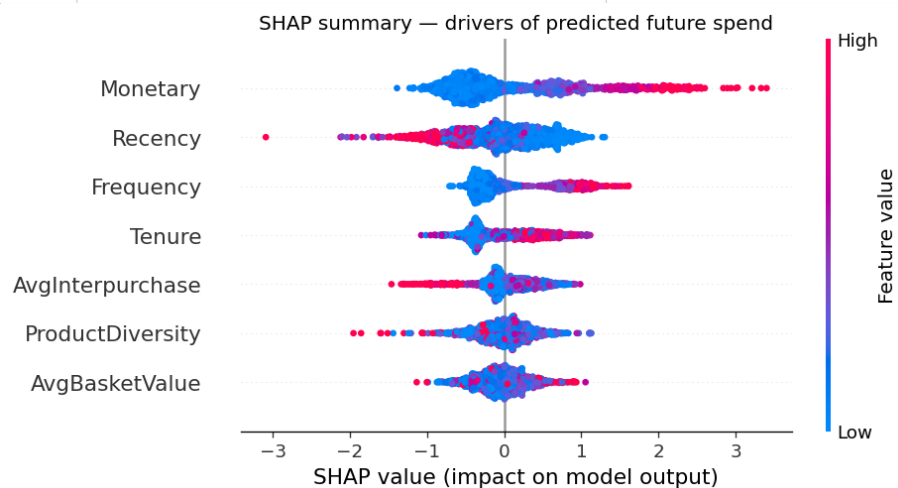


Figure 7. SHAP summary — Monetary, Recency and Frequency dominate. The model independently rediscovers RFM, validating the segmentation strategy.

Business value: trust in the AI, plus a clear targeting rule — focus retention budget on customers with high predicted CLV but rising Recency (high value, starting to slip away).

10. Business Value & Expected Impact

Lever	Mechanism	Expected impact
Cross-sell revenue	KNN + basket recommendations	+5-15% basket revenue
Customer retention	Segment-targeted win-back	Lower VIP/Loyal churn
Marketing efficiency	CLV-prioritised spend	Higher ROI per GBP
Product bundling	Apriori product pairs	Larger average basket
Inventory planning	Seasonality + top-product demand	Fewer stockouts/overstock

Deliverables

- Saved models: customer segmentation, KNN recommender, CLV (joblib).
- Per-customer top-10 recommendations and forward CLV scores.
- Power BI star schema (fact_sales + customer/product/date dimensions).
- All figures, metric tables, and this report.

Tech stack

Python • pandas • scikit-learn • XGBoost • LightGBM • mlxtend (Apriori) • SHAP • matplotlib/seaborn • Power BI.

This project demonstrates customer analytics, recommendation systems, clustering, association-rule mining, explainable AI and business intelligence within a single, reproducible, end-to-end retail solution.